

Homework 1

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Exercise 1

Part (a)

p	q	$\neg p \vee \neg q$	$p \vee q$	$(\neg p \wedge q) \vee (p \wedge \neg q)$
0	0	1	0	0
0	i	1	i	i
0	1	1	1	1
i	0	1	i	i
i	i	i	i	i
i	1	i	1	i
1	0	1	1	1
1	i	i	1	i
1	1	0	1	0

The argument is valid in both Strong Kleene logic and the Logic of Paradox. This does not, however, tell us anything about its validity in the Logic of First-Degree Entailment. The following derivation works for Strong Kleene logic.

1		$\neg p \vee \neg q$			
2		$p \vee q$			
3			p		
4				$\neg p$	
5				$(\neg p \wedge q) \vee (p \wedge \neg q)$	$\neg E3, 4$
6				$\neg q$	
7				$p \wedge \neg q$	$\wedge I3, 6$
8				$(\neg p \wedge q) \vee (p \wedge \neg q)$	$\vee I7$
9				$(\neg p \wedge q) \vee (p \wedge \neg q)$	$\vee E1, 4-5, 6-8$

10			q	
11			$\neg p$	
12			$\neg p \wedge q$	$\wedge I_{10, 11}$
13			$(\neg p \wedge q) \vee (p \wedge \neg q)$	$\vee I_{12}$
14			$\neg q$	
15			$(\neg p \wedge q) \vee (p \wedge \neg q)$	$\vee I_{10, 14}$
16			$(\neg p \wedge q) \vee (p \wedge \neg q)$	$\vee E_{1, 11-13, 14-15}$
17			$(\neg p \wedge q) \vee (p \wedge \neg q)$	$\vee E_{2, 3-9, 10-16}$

The following derivation works for the Logic of Paradox.

1		$\neg p \vee \neg q$			
2		$p \vee q$			
3			p		
4				q	
5				$\neg p$	
6				$\neg p \wedge q$	$\wedge I4, 5$
7				$(\neg p \wedge q) \vee (p \wedge \neg q)$	$\vee I6$
8				$\neg q$	
9				$p \wedge \neg q$	$\wedge I3, 8$
10				$(\neg p \wedge q) \vee (p \wedge \neg q)$	$\vee I9$
11				$(\neg p \wedge q) \vee (p \wedge \neg q)$	$\vee E1, 5-7, 8-10$
12				$\neg q$	
13				$p \wedge \neg q$	$\wedge I3, 12$
14				$(\neg p \wedge q) \vee (p \wedge \neg q)$	$\vee I13$
15				$(\neg p \wedge q) \vee (p \wedge \neg q)$	$TND4-11, 12-14$

16		<u>q</u>		
17		<u>p</u>		
18		$\neg p$		
19		$\neg p \wedge q$	$\wedge I16, 18$	
20		$(\neg p \wedge q) \vee (p \wedge \neg q)$	$\vee I19$	
21		$\neg q$		
22		$p \wedge \neg q$	$\wedge I17, 21$	
23		$(\neg p \wedge q) \vee (p \wedge \neg q)$	$\vee I22$	
24		$(\neg p \wedge q) \vee (p \wedge \neg q)$	$\vee E1, 18-20, 21-23$	
25		$\neg p$		
26		$\neg p \wedge q$	$\wedge I16, 25$	
27		$(\neg p \wedge q) \vee (p \wedge \neg q)$	$\vee I26$	
28		$(\neg p \wedge q) \vee (p \wedge \neg q)$	$TND17-24, 25-27$	
29		$(\neg p \wedge q) \vee (p \wedge \neg q)$	$\vee E2, 3-15, 16-28$	

Part (b)

p	q	$p \Rightarrow q$	$p \Rightarrow (p \wedge q)$
0	0	1	1
0	i	1	1
0	1	1	1
i	0	i	i
i	i	i	i
i	1	1	i
1	0	0	0
1	i	i	i
1	1	1	1

The argument is not valid in Strong Kleene logic. The valuation where $p = i$ and $q = 1$ is a counterexample. However, it is valid in the Logic of Paradox. Since the argument is not valid in Strong Kleene logic, it then cannot be valid in the Logic of First-Degree Entailment since it is weaker than Strong Kleene

logic. The following derivation works for the Logic of Paradox.

1	$p \Rightarrow q$	
2	$\neg p \vee q$	mcE1
3	$\neg p$	
4	$\neg p \vee (p \wedge q)$	$\vee I3$
5	$p \Rightarrow (p \wedge q)$	mcI4
6	q	
7	p	
8	$p \wedge q$	$\wedge I6, 7$
9	$\neg p \vee (p \wedge q)$	$\vee I8$
10	$p \Rightarrow (p \wedge q)$	mcI9
11	$\neg p$	
12	$\neg p \vee (p \wedge q)$	$\vee I11$
13	$p \Rightarrow (p \wedge q)$	mcI12
14	$p \Rightarrow (p \wedge q)$	TND7–10, 11–13
15	$p \Rightarrow (p \wedge q)$	$\vee E2, 3–5, 6–14$

Part (c)

p	q	$\neg(p \Rightarrow p) \vee \neg(q \Rightarrow p)$	p
0	0	0	0
0	i	i	0
0	1	1	0
i	0	i	i
i	i	i	i
i	1	i	i
1	0	0	1
1	i	0	1
1	1	0	1

The argument is not valid in either Strong Kleene logic or the Logic of Paradox. The valuation where $p = 0$ and $q = 1$ is a counterexample for both. Since the argument is not valid in either Strong Kleene logic or the Logic of Paradox, it then cannot be valid in the Logic of First-Degree Entailment since it is weaker than both Strong Kleene logic and the Logic of Paradox.

Exercise 2

Part (a)

1		$\neg A$	
2		$(\neg A \Rightarrow A) \vee \neg A$	$\vee I1$
3		$\neg A \Rightarrow A$	
4		$(\neg A \Rightarrow A) \vee \neg A$	$\vee I3$
5		$(\neg A \Rightarrow A) \vee \neg A$	$L_3 D1-2, 3-4$

We then see that $\vdash (\neg A \Rightarrow A) \vee \neg A$ is a valid argument in Łukasiewicz logic.

1		B	
2		A	
3		B	$R1$
4		$\neg B$	
5		$\neg A$	$\neg E1, 4$
6		$A \Rightarrow B$	$L_3 \Rightarrow I2-3, 4-6$

We then see that $B \vdash A \Rightarrow B$ is a valid argument in Łukasiewicz logic.

Part (b)

1	$\neg(A \vee B) \Rightarrow (A \vee B)$	
2	$(\neg A \Rightarrow A) \vee \neg A$	$T\Diamond$
3	$\neg A \Rightarrow A$	
4	$(\neg A \Rightarrow A) \vee (\neg B \Rightarrow B)$	$\vee I3$
5	$\neg A$	
6	$(\neg B \Rightarrow B) \vee \neg B$	$T\Diamond$
7	$\neg B \Rightarrow B$	
8	$(\neg A \Rightarrow A) \vee (\neg B \Rightarrow B)$	$\vee I7$
9	$\neg B$	
10	$\neg(A \vee B)$	$\neg\vee I5, 9$
11	$A \vee B$	$\Rightarrow E1, 10$
12	A	
13	$\neg A \Rightarrow A$	$W12$
14	$(\neg A \Rightarrow A) \vee (\neg B \Rightarrow B)$	$\vee I13$
15	B	
16	$\neg B \Rightarrow B$	$W15$
17	$(\neg A \Rightarrow A) \vee (\neg B \Rightarrow B)$	$\vee I17$
18	$(\neg A \Rightarrow A) \vee (\neg B \Rightarrow B)$	$\vee E11, 12-14, 15-17$
19	$(\neg A \Rightarrow A) \vee (\neg B \Rightarrow B)$	$\vee E6, 7-8, 9-17$
20	$(\neg A \Rightarrow A) \vee (\neg B \Rightarrow B)$	$\vee E2, 3-4, 5-19$

We then see that $\neg(A \vee B) \Rightarrow (A \vee B) \vdash (\neg A \Rightarrow A) \vee (\neg B \Rightarrow B)$ is a valid argument in Łukasiewicz logic.

Exercise 3

Part (a)

Let v, v' be LP-valuations such that $v \preceq v'$. Let $*$ be a connective such that $v(*A) = i$ regardless of A . Clearly, $v'(*A) \neq 0$ for all A ; it then follows that, vacuously, $v(*A) = 0$ only when $v'(*A) = 0$. Similarly, $v(*A) = 1$ only when $v'(*A) = 1$, and so Proposition 7.1.3 still holds if this connective is added to the Logic of Paradox.

Part (b)

Let $\#$ be a connective such that:

$$p\#q = \begin{cases} 0 & p, q = 0 \\ i & p = 0, q = i \text{ or } p = i, q = 0 \\ 1 & \text{otherwise} \end{cases}$$

Let v be a valuation where $v(p) = i$ and $v(q) = i$, and let v' be a valuation where $v'(p) = 0$ and $v'(q) = 0$. We see that $v \preceq v'$. However, $v(p\#q) = i\#i = 1$ while $v'(p\#q) = 0\#0 = 0$. Consequently, Proposition 7.1.3 no longer holds if this connective is added to the Logic of Paradox.